

6.2: THE PIGEONHOLE PRINCIPLE:

If K is a positive integer and $K+1$ or more objects are placed into K boxes, then there is at least one box containing two or more of the objects.

EXAMPLE #1:

Among the group of 367 people, there must be at least 2 with the same birthday.

Sol:

To make a challenge to pigeonhole principle, we may select all people in different date, even then

Sol: # of pigeonholes = # of days in an year = $K = 366$ (incl. leap year)
of pigeons = # of people = $K+1 = 367$

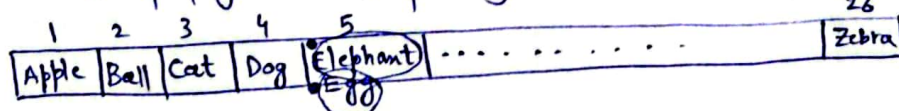
So, according to pigeonhole principle, there must be at least 2 people with the same birthday.

Example #2

In a group of 27 English words, there must be at least 2 that begin with the same letter.

Sol.

of pigeonholes = # of English letters = $K = 26$
of pigeons = # of English words = $27 = K+1$



Even After taking worst case i.e., we may select all the words in separate pigeonhole.

There must be at least two English words that begin with same letter.

Example #3

How many students must be in the class to guarantee that at least 2 students receive the same score. (Exam is graded on a scale 0-100 points)

Sol.

of pigeonholes = $K = 101$

To guarantee that at least 2 students receive the same score

$K+1$ pigeons are required to guarantee, so

$K+1 = 101+1 = 102$ # of students in a class

PIGEONHOLE principle is a useful tool in many proofs, ~~including~~ including proofs of surprising results such as that given in EXAMPLE 4 (kind of real problems). The above three examples are simple case of pigeonhole principle.

EXAMPLE #4 :

Show that for every n , there is a multiple of n that has only 0s and 1s in its decimal expansion.

3 x ? = (Decimal)
Containing only 0's and 1's

Sol:

Let n be a positive integer. Suppose $n=3$

Consider $n+1$ integers (i.e. $3+1=4$ integers), ~~where~~ ^{that} have only 1's in its decimal expansion } the reason

Four integers having only 1's :

1st. 1 one

2nd. 11 Eleven

3rd. 111 one hundred Eleven

4th. 1111 one thousand one hundred Eleven

the reason we choose more than one the number, because we want to apply pigeonhole principle

List of 4 integers = $\{1, 11, 111, 1111\}$

Every integer in the list when divided by n (i.e. divided by 3), there are n possible remainders (i.e. 3 possible remainders 0, 1 & 2)

integers : 1 11 111 1111
 remainders : ① 2 0 ①



 Same remainder

(we are guaranteed to have at least 2 remainder)
(will be same)

The basic idea is :
Since, there are 4 integers
then, if we divide each
by 3, at least two of
the remainders will be
the same and that
comes because of
pigeonhole principle

Because there are $n+1$ integers in the list,

By pigeonhole principle,

By pigeonhole principle,
There must be two integers (having only 1's in its decimal expansion) with the same remainder when divided by n .

Multiple of n

The Larger of these integers less the Smaller one is a multiple of n , which has a decimal expansion consisting of entirely of 0's and 1's.

largest = 1111
smallest = 1
(with same remainder)

$$\begin{array}{r} 111 \\ - 1110 \\ \hline \end{array}$$

$$3 \times ? = 1110$$

$$3 \times 370 = 1110$$

$$\begin{array}{r} 370 \\ 3 \overline{) 1110} \\ \underline{9} \\ 21 \\ \underline{21} \\ 0 \end{array}$$

Let $n=4$

five integers $= n+1 = \{1, 11, 111, 1111, 11111\}$

remainder 1 3 3 3 3

largest and smallest with same remainder

$a = 11111$ $b = 11$

Largest - Smallest $= 11111 - 11 = 11100$ $n|a-b$ i.e. $3|a-b$

Let n is an integer, so, there is a multiple of n that has only 0's and 1's, in its decimal expansion.

$4 * ? = 11100$

$4 * 2775 = 11100$

$$\begin{array}{r} 2775 \\ 4 \overline{) 11100} \\ \underline{8} \\ 31 \\ \underline{28} \\ 30 \\ \underline{28} \\ 20 \end{array}$$

GENERALIZED PIGEONHOLE PRINCIPLE

If N objects are placed into K boxes, then there is at least one box containing at least $\lceil \frac{N}{K} \rceil$ objects.

Note: A common type of problem asks for the minimum # of objects such that at least r of these objects must be in the K boxes when these objects are distributed among the boxes.

Normal Pigeonhole principle:

It would just guarantee for at least 2 with the same pigeonhole.

Generalized Pigeonhole Principle:

Generalized pigeonhole principle tells us more than the normal pigeonhole principle.

In generalized principle, we cannot just guarantee at least 2 with the same pigeonhole but we can guarantee ^{at least} 3 or 4 or 5 with the same pigeonhole.

EXAMPLE 1:

a) What is the minimum number of people such that at least 9 of them have same month of birth.

Sol:-

$r = 9$ (at least pigeons in one box)

$K = 12$ (months i.e. # of pigeonholes)

$N = \#$ of people (min. # of people)

$$r-1 < \frac{N}{K} \leq r$$

$$8 < \frac{N}{12} \leq 9$$

$$8 \times 12 < N \leq 9 \times 12$$

$$96 < N \leq 108$$

1	2	3	4	...	11	12
8	8	8	8	...	8	9
$11 \times 8 = 88$						9
						$88 + 9 = 97$

Range of N :

the range of people such that at least 9 of them have same birth month is greater than 96 and less than or equal to 108.

$$N = K(r-1) + 1 \\ = 12(9-1) + 1 = 12 \times 8 + 1 = 97$$

Smallest integer N

$$\frac{N}{K} > r-1$$

$$\text{or } N > K(r-1)$$

$$\text{or } N = K(r-1) + 1$$

EXAMPLE #1

b) Among 100 people, what is the least #, who were born in same month.

Sol: $N = 100$ (# of pigeons)
 $K = 12$ (# of pigeonholes)
 $r = ?$ (at least r ~~have~~ ^{were born} in same month)

$$\left\lceil \frac{N}{K} \right\rceil = r$$

$$\left\lceil \frac{100}{12} \right\rceil = \lceil 8.33 \rceil = 9$$

1	2	3	3	4		9	10	11	12
8	8	8	8	8	..	9	9	9	9
$8 \times 8 = 64$						$9 \times 4 = 36$			
$64 + 36 = 100$									

EXAMPLE #2:

What is the minimum # of students required in a discrete structure class that at least 6 will receive the same grade, if there are possible 5 grades, A, B, C, D, F.

Sol: $r = 6$ (at least pigeons in one box)
 $K = 5$ (# of pigeonholes)
 # of pigeons = $N = ?$ (minimum N)

$$r - 1 < \frac{N}{K} \leq r$$

$$5 < \frac{N}{5} \leq 6$$

$$5 \times 5 < N \leq 6 \times 5$$

$$\text{+1: } \min(25) < N \leq \max(30)$$

Range of N (i.e. range of students):
 greater than 25
 Less than or equal to 30

A	B	C	D	F
5 dots	5 dots	5 dots	5 dots	6 dots
5	5	5	5	6
≈ 26 (min)				

6	6	6	6	6
$= 30$ (max)				

Smallest N :

$$\frac{N}{K} > r - 1 \Rightarrow N = K(r - 1) + 1$$

$$N = 5 \times 5 + 1 = 26 \text{ min. \# of students}$$

$$N = K(r - 1) + (r - 1) \text{ max. \# of students}$$

$$= 5 \times 5 + 5 = 25 + 5 = 30$$

EXAMPLE #3

- a) How many cards must be selected from a deck of 52 cards to guarantee that at least 3 cards of the same suit are chosen?
- b) How many must be selected to guarantee that at least 3 hearts are selected.

Sol:-

a) ~~Heart~~

1	2	3	4
Heart	Spades	Clubs	Diamond

$K = 4$ Suits (# of pigeonholes)

$r = 3$ (at least 3 cards of the same suit)

$N = ?$ (# of pigeons)

$$2 < \frac{N}{4} \leq 3$$

$$2 \times 4 < N \leq 3 \times 4$$

$$8 < N \leq 12$$

The smallest # of cards

$$N = K(r-1) + 1$$

$$N = 4(2) + 1 = 9$$

- b) We do not use generalized pigeonhole principle here to answer this, because we want to make sure that there are 3 hearts, not just 3 cards of one suit.

We may select all Clubs, diamonds and spades before any heart i.e., 39 Cards.

The next 3 Cards will be all hearts, so we may need to select $39 + 3 = 42$ Cards to get 3 hearts.

Example #4

What is the minimum # of students in the class such that

- a) There must be at least 2 with same birthday (day of week).
- b) " " " " " 3 " " " " " " " "

Sol:

a) $K = 7$ (Days of week i.e. pigeonholes)

$r = 2$ $(r-1) < \frac{N}{K} \leq r$

$N = ?$ $1 < \frac{N}{7} \leq 2$

$$7 < N \leq 14$$

Smallest

$$N = K(r-1) + 1$$

$$N = 7 \times 1 + 1 = 8$$

1	2	3	4	5	6	7
.	.	.	.	.	.	.
1	1	1	1	1	1	2

$$\left\lceil \frac{8}{7} \right\rceil = 2$$

$$\left\lceil \frac{9}{7} \right\rceil = 2$$

...

$$\left\lceil \frac{14}{7} \right\rceil = 2$$

$$\left\lceil \frac{15}{7} \right\rceil = 3$$

$$\left\lceil \frac{16}{7} \right\rceil = 3$$

$$\left\lceil \frac{21}{7} \right\rceil = 3$$

$$\left\lceil \frac{8 \rightarrow 14}{7} \right\rceil = 2$$

$$\left\lceil \frac{15 \text{ to } 21}{7} \right\rceil = 3$$

b)

$K = 7$

$r = 3$

$(r-1) < \frac{N}{K} \leq r$

$2 < \frac{N}{7} \leq 3$

$$14 < N \leq 21$$

$$N = K(r-1) + 1$$

$$N = 7(2) + 1 = 15$$